NASA INTERNATIONAL SPACE APPS CHALLENGE 2026 | THEME: THE NEXT FRONTIER

MASTER VIDEO SCRIPT & PRODUCTION BLUEPRINT

RELICS OF THE COSMIC DUST

Interactive 3D Planetary Archaeology & STEM Storytelling Platform

Challenge: Abandoned but Not Forgotten: NASA's Discarded Equipment on Moon & Mars

Exact Video Runtime: 240 Seconds (04:00 Minutes)

Team Name: TEAM RELICS

Technology Core: Zero-Install WebGL (Three.js) & Web Audio API

01 Team Members & Production Roles

TEAM MEMBER	PRODUCTION ROLE	CORE VIDEO & ENGINEERING RESPONSIBILITIES
Shaikot Kundu Akash	Team Lead	Overall project coordination, system architecture, lead presenter VO, repository oversight.
Md. Wahiduzzaman Nayem	System Analyst & Video Editor	Video editing, motion graphics, audio sync, classroom shot pacing, data architecture review.
Arni Debnath	Lead Designer	UI/UX wireframing, dark glassmorphism design system, infographics, certificate rendering.
Tanjir Ahmed Fardin	Researcher & Storyteller	NASA PDS & mission literature review, narrative scriptwriting, educational dialogue design.
Tareq Jamil Sarkar Samrat	Frontend & Mechanics Developer	Three.js 3D physics engine, rover kinematics, Sol 5111 atmospheric shaders, STEM quiz mechanics.

02 Master NASA Open Data & Scientific Pipeline Ingestion

Our platform directly integrates verified NASA open datasets to drive real-time 3D terrain generation, physics-based atmospheric extinction calculations, authentic CAD geometry, and historical telemetry:

NASA DATA SOURCE	MISSION & SCIENTIFIC DOMAIN	RAW FORMAT	DIRECT IMPLEMENTATION IN SOLUTION
NASA PDS Geosciences Node	MER-B Opportunity (Sols 1–5111)	ASCII / PDS3 Tables	Historical atmospheric tau logs (0.8–10.8), power collapse (900 Wh to 22 Wh), 20.5V battery cutoff, Beer-Lambert solar extinction formula: `I = I₀ · e^(-τ / cos θ)`.
NASA MRO HiRISE Archive	Mars Reconnaissance Orbiter	16-bit GeoTIFF DTM	Sub-meter elevation heightmap for Perseverance Valley at Endeavour Crater (-14.57°S, 354.67°E).
NASA LRO LOLA Altimetry	Lunar Reconnaissance Orbiter	16-bit PDS IMG / GeoTIFF	Laser altimeter DTM elevation maps for historical lunar sites including Oceanus Procellarum (2.47°S, 316.66°E).
NASA 3D Resources & JPL	Surveyor 1 & MER-B Opportunity	.obj / .gltf 3D Meshes	Authentic 3D rover and lander models with functional 6-wheel rocker-bogie suspension physics.
NASA Mineralogical Archives	APXS, Mössbauer & RAT Logs	Spectral CSV / PDS Data	Aqueous mineralogy: Hematite Blueberries at Eagle Crater, Jarosite identification, and rock abrasion grind-hole telemetry.

⚡ **DIRECT DATA PIPELINE FLOW**

Raw NASA PDS / GeoTIFF Data ➔ GDAL Raster Converter ➔ 16-Bit Grayscale Heightmap Grids ➔ Three.js Procedural WebGL Terrain + Ray-marched Atmospheric Dust Shaders.

03 240-Second (4-Minute) Master Production Script

BLOCK 1: INTRODUCTION & TEAM REVEAL

0:00 – 0:30 (30 Seconds)

SCENE 1: Establishing the Smart Classroom & Team Reveal

0:00 – 0:10

VISUAL Wide establishing shot of a modern 2026 smart classroom full of students. Morning sunlight streams through large windows; front monitor is asleep. Mr. Kabir stands near his desk smiling. At 0:05, a sleek semi-transparent glassmorphic overlay appears with the NASA Space Apps 2026 badge and Team Relics crest.

ON-SCREEN **TEAM RELICS:** Shaikot Kundu Akash (Lead), Md. Wahiduzzaman Nayem (Systems & Video), Arni Debnath (Design), Tanjir Ahmed Fardin (Research & Story), Tareq Jamil Sarkar Samrat (Frontend & Core Mechanics).

AUDIO & SFX Classroom murmur fades into an energetic synth pulse. Smooth digital UI chime.

> **VO (VOICE-OVER, 0:05 – 0:12)**
> *"We are Team Relics—led by Shaikot, with systems and editing by Nayem, UI/UX design by Arni, research and storytelling by Fardin, and interactive mechanics built by Samrat."*

SCENE 2: The Student's Question & The Next Frontier

0:10 – 0:30

VISUAL Medium close-up on Emon, a curious 12-year-old boy in a striped t-shirt raising his hand eagerly. Cut to wide shot at 0:20 with lower-third displaying the NASA 2026 challenge title.

ON-SCREEN *NASA Space Apps 2026 | Theme: The Next Frontier | Challenge: Abandoned but Not Forgotten*

AUDIO & SFX Classroom hush, single chair creak, ambient synth continues low in mix.

> **EMON (IN-SCENE DIALOGUE, 0:10 – 0:18)**
> *"Sir! NASA sent so many rovers and landers to the Moon and Mars, right? What happened to them? Did they even complete their missions?"*

> **VO (NARRATOR, 0:19 – 0:30)**
> *"Under the 2026 Space Apps theme, 'The Next Frontier', we set out to answer Emon's question by tackling the challenge: Abandoned but Not Forgotten: Storytelling About NASA's Discarded Equipment on the Moon and Mars."*

BLOCK 2: THE PROBLEM — INACCESSIBLE COLD ARCHIVES

0:30 – 1:00 (30 Seconds)

SCENES 3, 4 & 5: Teacher's Search & The Data Wall

0:30 – 0:48

VISUAL Mr. Kabir smiles warmly and types N-A-S-A on a mechanical keyboard with glowing ripples. The giant classroom monitor fills with a dense, text-heavy space archive of raw catalog numbers and tiny thumbnails.

AUDIO & SFX 4 distinct mechanical key-clacks, enter chime, rapid mouse scrolling sounds.

> **MR. KABIR (IN-SCENE DIALOGUE, 0:30 – 0:46)**
> *"Great question, Emon! Let's find out together—let me pull up NASA's official mission archive... let's scroll down and see."*

SCENES 6 & 7: Boring Deep Dive & Students Losing Interest

0:48 – 1:00

VISUAL	Close-up of endless scrolling tables. Panning shot across the room: students slumping over desks, one resting head on arms, another doodling.
ON-SCREEN	Highlight box over unreadable telemetry: *[Table 4.1: MER-B PDS Raw Binary Telemetry Logs — Sol 5111]*
AUDIO & SFX	Clock ticking softly, exaggerated student yawn, pencil scratch on paper.

> **MR. KABIR (MUTTERING, 0:48 – 0:52)**
> *"Hmm... so many raw tables... where is the story behind all this?"*

> **VO (NARRATOR, 0:52 – 1:00)**
> *"Decades of planetary exploration have left monumental hardware across the Moon and Mars. Yet, trapped inside dry, complex archives, these triumphs feel like forgotten space junk—draining classroom curiosity before it can start."*

BLOCK 3: NASA DATA SOURCES & INGESTION PIPELINE

`1:00 - 1:30 (30 Seconds)`

SCENES 8, 9 & 10: Emon's Spark & NASA Open Data Architecture

`1:00 - 1:18`

VISUAL Emon sits upright with a spark in his eye and hands Mr. Kabir a URL note. Mr. Kabir enters the link. A high-tech split-screen schematic reveals NASA's open-data architecture feeding into the platform.

ON-SCREEN
- 🔴 **NASA PDS:** Sols 1–5111 Telemetry & Atmospheric Tau
- 🟡 **NASA LRO LOLA:** 16-Bit Lunar Digital Elevation Models (DTM)
- 🪶 **NASA MRO HiRISE:** Perseverance Valley Sub-Meter Topographical Grids
- 🛰️ **NASA 3D Resources / JPL:** Official CAD Hardware Geometry

AUDIO & SFX Idea chime, rapid keyboard clacks, futuristic UI processing hum.

> **EMON (IN-SCENE, 1:08 – 1:12)**
> *"Sir, try this! It turns real NASA mission data into a 3D world!"*

> **VO (NARRATOR, 1:12 – 1:18)**
> *"To solve this, we tapped directly into NASA Open Data. We transformed high-resolution digital elevation models from the Lunar Reconnaissance Orbiter and Mars HiRISE..."*

PIPELINE TRANSITION: Direct Data Pipeline & Scientific Rigor

`1:18 - 1:30`

VISUAL Animated schematic: Raw NASA GeoTIFF / PDS3 ➔ GDAL Processing ➔ 16-Bit Grayscale Heightmaps ➔ Three.js Procedural Terrain Engine with Beer-Lambert Law physics dynamically calculating sunlight drop.

ON-SCREEN *Data Pipeline: NASA PDS ➔ GDAL ➔ Three.js Procedural Engine | Physics:* $I = I_0 \cdot \exp(-\tau / \cos\theta)$

AUDIO & SFX Rising musical tension, high-tech sweep effect.

> **VO (NARRATOR, 1:18 – 1:30)**
> *"...coupling them with NASA's Planetary Data System telemetry and official 3D hardware assets to recreate authentic, scientifically accurate extraterrestrial environments directly in the browser."*

BLOCK 4: THE SOLUTION & LIVE INTERACTIVE SIMULATION

`1:30 - 3:30 (120 Seconds / 2 Minutes)`

SCENES 11 & 12: Platform Reveal & Instant Browser Loading

`1:30 - 1:55`

VISUAL The classroom monitor explodes with vibrant color: dark space web app with title "RELICS OF THE COSMIC DUST" glowing in gold and cyan. Students gasp in awe. Mr. Kabir clicks [EXPLORE HISTORIC SITES].

ON-SCREEN *RELICS OF THE COSMIC DUST — Zero-Install WebGL Platform (Live in Any Browser)*

AUDIO & SFX Orchestral swell, particle whoosh, crisp UI click.

> **STUDENTS (1:35 – 1:40)**
> *"Whoa!" / "Look at the graphics!"*

> **MR. KABIR (1:40 – 1:44)**
> *"Now THIS looks incredible. Look at that detail!"*

> **VO (NARRATOR, 1:44 – 1:55)**
> *"Meet 'Relics of the Cosmic Dust'—an ultra-lightweight, zero-install WebGL interactive simulation that loads in under three seconds on any classroom device without downloads or setup."*

SCENES 13 & 14: 3D Lunar Simulator & Driving the Rover

1:55 – 2:25

VISUAL Low-gravity dust drifts across Oceanus Procellarum. In distance, a golden beacon pulses. Split shot: students pressing WASD keys while the 3D rover traverses grey lunar soil kicking up dust puffs.

ON-SCREEN *Location: Oceanus Procellarum (2.47°S, 316.66°E) | Elevation: NASA LRO LOLA Altimetry | Live HUD: Coordinates & Incline*

AUDIO & SFX Ambient lunar hum, key clicks, electric rover motor whirr, students cheering ("Go, turn left!").

> **MR. KABIR (2:00 – 2:06)**
> *"Look—that beacon marks a real lunar landing site from 1966!"*

> **VO (NARRATOR, 2:06 – 2:25)**
> *"Students step directly into the shoes of Planetary Archaeologists. Navigating authentic lunar topography generated from NASA elevation data, players use intuitive controls to track real telemetry beacons across historical exploration zones."*

SCENES 15 & 16: Proximity Arrival, LiDAR Scan & The Living Story

2:25 - 2:55

VISUAL The rover pulls up next to Surveyor 1. A holographic blue LiDAR scanning circle sweeps across the lander. An illustrated story modal unfolds with archival photos and engineering schematics.

ON-SCREEN *Target: Surveyor 1 (Landed: June 2, 1966) | Achievement: First U.S. Soft Lunar Touchdown | Proved soil firm for Apollo*

AUDIO & SFX Holographic scan sweep, electronic chime, gentle emotional string music.

> **MR. KABIR (2:27 – 2:30)**
> *"Now let's scan its systems and uncover its story."*

> **IN-GAME NARRATOR / VO (2:35 – 2:55)**
> *"June 1966. Surveyor 1 softly settled into the lunar dust, proving the ground was firm enough for human boots. It was forgotten for decades—but its solitary landing paved the way for every human step on the Moon."*

SCENES 17, 18 & 19: Gamified STEM Quiz & Martian Sol 5111 Re-enactment

2:55 - 3:30

VISUAL STEM quiz card appears: "What did Surveyor 1 prove about lunar soil?" Students shout answers. Correct choice clicked ➔ Confetti pop and XP surge. Camera transitions to Mars (Perseverance Valley): sky turns dark red as PDS atmospheric tau hits 10.8, showing Opportunity MER-B's final Sol 5111 storm.

ON-SCREEN *Achievement Unlocked: Lunar Pioneer (+250 STEM XP)*
Mars Ground Truth: Perseverance Valley (-14.57°S, 354.67°E | Sol 5111 Tau=10.8)
PDS Telemetry: Power collapse 900 Wh ➔ 22 Wh/sol (Cutoff: 20.5V)

AUDIO & SFX Quiz chime, confetti pop, classroom cheering, procedural Martian storm wind, upbeat music.

> **STUDENTS (2:58 – 3:02)**
> *"Option B! The soil was firm!"*

> **MR. KABIR (3:02 – 3:06)**
> *"Correct! Brilliant teamwork, everyone!"*

> **VO (NARRATOR, 3:06 – 3:30)**
> *"Every relic scanned connects history with gamified STEM learning. Powered by real PDS telemetry, students witness the exact conditions that silenced Opportunity on Sol 5111, test their knowledge with quizzes, and unlock badges on their journey to becoming Master Planetary Archaeologists."*

BLOCK 5: EDUCATIONAL IMPACT & OPEN SCIENCE FINALE

3:30 - 4:00 (30 Seconds)

SCENE 20: Classroom Celebration & Global Call-to-Action

3:30 – 4:00

VISUAL
Wide shot of classroom giving standing ovation. Mr. Kabir smiles proudly beside the golden "Master Planetary Archaeologist" certificate. Camera tilts out through the window into the sky showing Moon and Mars. Project title card, live URL, and GitHub QR code fade in.

ON-SCREEN
RELICS OF THE COSMIC DUST | Developed by TEAM RELICS
Live Simulator: [Live-Build-URL] | Open-Source Pipeline: [GitHub-URL]
NASA International Space Apps Challenge 2026

AUDIO & SFX
Triumphant orchestral finale swelling to emotional peak; classroom applause gently fades.

> **MR. KABIR (3:32 – 3:38)**
> *"Look at that—history isn't boring at all, is it? Sometimes you just need the right way to explore it."*

> **VO (NARRATOR / TEAM LEAD SHAIKOT, 3:38 – 4:00)**
> *"NASA's hardware may sit abandoned across deep space, but their legacy will never be forgotten. By turning cold data into living stories, Team Relics is inspiring the next generation of planetary explorers. Experience the live simulator right now in your browser. Thank you!"*

04 Master Reference & Technical Pacing Table

TIME WINDOW	SCRIPT SECTION	SCENES	PRIMARY VISUAL & NASA CONTENT	AUDIO STATE
0:00 – 0:30	Intro & Team	Scene 1, 2	Classroom wide shot; Team Relics roster; Emon's question; 2026 Space Apps challenge theme.	Ambient chatter ➔ Synth pulse ➔ Focused dialogue.
0:30 – 1:00	The Problem	Scene 3 – 7	NASA archive search; raw PDS tables; students losing interest and slumped over desks.	Key-clacks ➔ Mouse scroll ➔ Yawn / clock tick ➔ Narrative VO.
1:00 – 1:30	Data Pipeline	Scene 8 – 10	Emon's note; NASA open data architecture: LOLA, HiRISE, PDS logs, and GDAL to WebGL pipeline.	Idea chime ➔ Clacks ➔ Rising high-tech synth.
1:30 – 2:25	Solution & 3D Drive	Scene 11 – 14	Zero-install WebGL load in <3s; driving 3D rover on Moon across Oceanus Procellarum.	Orchestral swell ➔ UI clicks ➔ Rover motor hum.
2:25 – 3:30	Scan & Gamification	Scene 15 – 19	Scanning Surveyor 1; story modal; STEM quiz; badge unlock; transition to Mars Sol 5111 storm.	Holographic sweep ➔ Emotional strings ➔ Confetti & cheering.
3:30 – 4:00	Impact & Outro	Scene 20	Standing ovation; Master Archaeologist certificate; tilt to sky; Live link & GitHub QR.	Triumphant orchestral climax ➔ Final closing statement.

05 NASA Scientific Open Data Citations

All data and historical records utilized in the project and video are verified open scientific resources:

- **NASA Planetary Data System (PDS) Geosciences Node:** MER-B Atmospheric Opacity & Telemetry Logs (https://pds-geosciences.wustl.edu/missions/mer/)
- **NASA Mars Reconnaissance Orbiter (MRO) HiRISE Archive:** Perseverance Valley DTM (https://www.uahirise.org/)
- **NASA Lunar Reconnaissance Orbiter (LRO) LOLA Altimeter:** Lunar Topography Archive (https://pds.nasa.gov/)
- **NASA 3D Resources & Jet Propulsion Laboratory:** Surveyor 1 & MER-B 3D Structural Assets (https://nasa3d.arc.nasa.gov/)
- **NASA JPL Mission Status Logs:** Opportunity Sol 1 to Sol 5111 End-of-Mission Report (https://mars.nasa.gov/mer/)

RELICS OF THE COSMIC DUST — Developed by **TEAM RELICS** for the **NASA International Space Apps Challenge 2026**

"Abandoned across deep space, but their legacy will never be forgotten."